

CODETANTRA

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5.1.1. Stacked Plot

21:39

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

7'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
8'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
9}
10df = pd.DataFrame(data)
11
12plt.stackplot(df['Month'], df['City_A_Temperature'],
13df['City_B_Temperature'], df['City_C_Temperature'])
14
15plt.xlabel('Month')
16plt.ylabel('Temperature')
17
18plt.title('Temperature Variation')

Average time1.958 s1958.00 msMaximum time1.958 s1958.00 ms1 out of 1 shown test case(s) passed

TerminalTest cases

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5.2.1. Titanic Dataset

13:58

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

titanicDat...

6
7# Set up the figure for 5 subplots
8fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9# write the code..
10count=df['Pclass'].value_counts().sort_index()
11count_gen=df['Gender'].value_counts()
12count_s=df['Survived'].value_counts().sort_index()
13
14axes[0,0].bar(count.index,count.values,color=['skyblue'])
15axes[0,0].set_title('Passenger Class Distribution')
16axes[0,0].set_xlabel('Pclass')
17axes[0,0].set_ylabel('Count')
18
19axes[0,1].pie(count_gen.values,labels=['male','female'],colors=['lightblue','lightcoral'],autopct='%1.1f%%')

Average time1.253 s1253.00 msMaximum time1.253 s1253.00 ms1 out of 1 shown test case(s) passed

TerminalTest cases

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

Sample Data:

Histogram...

```
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0],
10                           inplace=True)
11 data.drop('Cabin', axis=1, inplace=True)
12
13 # Convert categorical features to numeric
14 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
15 data = pd.get_dummies(data, columns=['Embarked'],
16                           drop_first=True)
17
18 # Write your code here for Histogram
19 plt.hist(data['Age'], bins=30, edgecolor='k')
```

Average time

0.370 s

370.00 ms

Maximum time

0.370 s

370.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the **'Survived'** column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to **'bar'**
3. Add the title **"Survival Count"** to the chart.
4. Label the x-axis as **'Survived'** and the y-axis as **'Count'**

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

BarPlotOf...

```
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'],
15                           drop_first=True)
16
17 # Write your code here for Bar Plot for Survival Rate
18 survival_counts = data['Survived'].value_counts()
19
20 survival_counts.plot(kind='bar')
21
22 plt.title('Survival Count')
23
24 plt.xlabel('Survived')
```

Average time

0.417 s

417.00 ms

Maximum time

0.417 s

417.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

BarPlotOf...

```

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 # Write your code here for Bar Plot for Survival by Gender
18
19 survival_by_gender = data.groupby('Sex')
20 ['Survived'].value_counts().unstack().fillna(0)
21 survival_by_gender.columns = ['Not Survived', 'Survived']
22 survival_by_gender.index = ['0', '1']
23 survival_by_gender.plot(kind='bar', stacked=True)
24 plt.title('Survival by Gender')
25 plt.xlabel('Gender')
26 plt.ylabel('Count')

```

Average time

0.521 s
521.00 ms

Maximum time

0.521 s
521.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Pclass"** to the chart.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

BarPlotOf...

```

14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 # Write your code here for Bar Plot for Survival by Pclass
18
19 survival_by_class = data.groupby('Pclass')
20 ["Survived"].value_counts().unstack().fillna(0)
21 survival_by_class.columns = ['Not Survived', 'Survived']
22 survival_by_class.plot(kind='bar', stacked=True)
23 plt.title('Survival by Pclass')
24 plt.xlabel('Pclass')
25 plt.ylabel('Count')
26 plt.legend(title=None)
27 plt.show()

```

Average time

0.419 s
419.00 ms

Maximum time

0.419 s
419.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.6. Bar Plot for Survival by Embarked

01:40

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked" to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

BarPlotOf...

```

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 # Write your code here for Bar Plot for Survival by Embarked
18
19 grouped = data.groupby('Embarked_Q')
20 ['Survived'].value_counts().unstack().fillna(0)
21 grouped.columns = ['Not Survived', 'Survived']
22 grouped.plot(kind='bar', stacked=True)
23 plt.title('Survival by Embarked')
24 plt.xlabel('Embarked')
25 plt.ylabel('Count')
26 plt.legend(title=None)
27 plt.show()

```

Average time

0.514 s

514.00 ms

Maximum time

0.514 s

514.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.7. Box plot for Age Distribution

00:13

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to "Age by Pclass".
3. Remove the default subtitle with `plt.suptitle('')`.
4. Label the x-axis as 'Pclass' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

Sample Data:

BoxPlotF...

```

14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 # Write your code here for Box Plot for Age by Pclass
18
19 plt.figure(figsize=(8, 6))
20
21 data.boxplot(column='Age', by='Pclass')
22
23 plt.suptitle('')
24
25 plt.title('Age by Pclass')
26
27 plt.xlabel('Pclass')
28
29 plt.ylabel('Age')
30
31 plt.show()

```

Terminal

Test cases

< Prev Reset Submit Next >

5.2.8. Box Plot for Age by Survived

03:39

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

BoxPlotF...

```
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 # Write your code here for Box Plot for Age by Survived
18
19 plt.figure(figsize=(8, 6))
20 data.boxplot(column='Age', by='Survived')
21 plt.suptitle('')
22 plt.title('Age by Survival')
23 plt.xlabel('Survived')
24 plt.ylabel('Age')
25 plt.show()
```

Average time

0.408 s

Maximum time

0.408 s

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.9. Box Plot for Fare by Pclass

01:45

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

BoxPlotF...

```
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 # Write your code here for Box Plot for Fare by Pclass
18
19 plt.figure(figsize=(8, 6))
20 data.boxplot(column='Fare', by='Pclass')
21 plt.suptitle('')
22 plt.title('Fare by Pclass')
23 plt.xlabel('Pclass')
24 plt.ylabel('Fare')
25 plt.show()
```

Average time

0.440 s

Maximum time

0.440 s

1 out of 1 shown test case(s) passed

Terminal

Test cases

< Prev Reset Submit Next >

5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

AgeFareS...

```

14 data = pd.get_dummies(data, columns=['Embarked'],
15 drop_first=True)
16
17 plt.figure(figsize=(6.4,4.8))
18
19 plt.scatter(data['Age'], data['Fare'])
20
21 plt.title('Age vs. Fare')
22
23 plt.xlabel('Age')
24
25 plt.ylabel('Fare')
26
27 plt.show()

```

Average time

0.379 s

379.00 ms

Maximum time

0.379 s

379.00 ms

1 out of 1 shown test case(s) passed

Test case 1

379 ms

Debug

Terminal

Test cases

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5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

AgeFareS...

```

9 data['Embarked'].fillna(data['Embarked'].mode()[0],
10 inplace=True)
11 data.drop('Cabin', axis=1, inplace=True)
12
13 # Convert categorical features to numeric
14 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
15 data = pd.get_dummies(data, columns=['Embarked'],
16 drop_first=True)
17
18 # Write your code here for Scatter Plot for Age vs. Fare by
19 Survived
20 colors = data['Survived'].map({0: 'red', 1: 'blue'})
21 plt.scatter(data['Age'], data['Fare'], c=colors)
22 plt.title('Age vs. Fare by Survival')
23 plt.xlabel('Age')
24 plt.ylabel('Fare')
25 plt.show()

```

Average time

0.408 s

408.00 ms

Maximum time

0.408 s

408.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

Prev Reset Submit